

that of $\Pr[1 \leftarrow \text{Dis}(x) : x \leftarrow \mathcal{D}(0)]$ ($\Pr[1 \leftarrow \text{Dis}(x) : x \leftarrow \mathcal{D}(1)]$). The reason why this learning algorithm works can be easily understood from the inequality

$$|\Pr[1 \leftarrow \text{Dis}(x) : x \leftarrow \mathcal{T}] - \Pr[1 \leftarrow \text{Dis}(x) : x \leftarrow \mathcal{D}(b)]| \leq \text{SD}(\mathcal{T}, \mathcal{D}(b)). \quad (3)$$

This learning algorithm can be implemented with access to an **NP** oracle, because Dis is the algorithm that takes x as input and outputs b if x is more likely from $\mathcal{D}(b)$, and such an algorithm can be implemented with the Stockmeyer theorem [Sto83]: a PPT algorithm with access to an **NP** oracle can approximate classical probability distributions. Hence, in this simplified setting, a PPT algorithm with access to an **NP** oracle can achieve agnostic distribution learning.

Next let us consider the actual case where $\mathcal{D}$'s input z is an n -bit string. Again, from the definition of the statistical distance, there exists an unbounded algorithm Dis such that

$$|\Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{D}(a)] - \Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{D}(b)]| = \text{SD}(\mathcal{D}(a), \mathcal{D}(b)) \quad (4)$$

for all a and b . Given $x_1, \dots, x_t$ independently sampled from $\mathcal{T}$, the learner first estimates the value of $\Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{T}]$ for all a, b . The learner outputs a^* such that

$$|\Pr[1 \leftarrow \text{Dis}(a^*, b, x) : x \leftarrow \mathcal{T}] - \Pr[1 \leftarrow \text{Dis}(a^*, b, x) : x \leftarrow \mathcal{D}(a^*)]| \quad (5)$$

is small for all b . The reason why this learning algorithm works can be understood from the following inequality:

$$\text{SD}(\mathcal{T}, \mathcal{D}(a)) \leq \text{SD}(\mathcal{T}, \mathcal{D}(b)) + \text{SD}(\mathcal{D}(a), \mathcal{D}(b)) \quad (6)$$

$$= \text{SD}(\mathcal{T}, \mathcal{D}(b)) + |\Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{D}(b)] - \Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{D}(a)]| \quad (7)$$

$$\leq \text{SD}(\mathcal{T}, \mathcal{D}(b)) + |\Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{D}(b)] - \Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{T}]| \quad (8)$$

$$+ |\Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{T}] - \Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{D}(a)]| \quad (9)$$

$$\leq 2\text{SD}(\mathcal{T}, \mathcal{D}(b)) + |\Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{T}] - \Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{D}(a)]|, \quad (10)$$

which suggests that in order to minimize $\text{SD}(\mathcal{T}, \mathcal{D}(a))$,

$$|\Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{T}] - \Pr[1 \leftarrow \text{Dis}(a, b, x) : x \leftarrow \mathcal{D}(a)]| \quad (11)$$

has to be minimized over a .

Finally, with a careful investigation, we show that the learning algorithm can be implemented with access to a Σ_3^{P} oracle.⁶

Proof of Theorem 1.5. In the following, we explain our proof of Theorem 1.5, which is **SampBQP** $\neq$ **SampBPP** from hardness of agnostic quantum distribution learning with respect to the statistical distance against $PPT^{\Sigma_3^{\text{P}}}$ learner. Let $\mathcal{T}$ be an unknown algorithm that outputs classical bit strings, and let $\mathcal{D}$ be a known QPT algorithm that takes a bit string z as input and outputs a classical bit string x . The learner who receives $x \leftarrow \mathcal{T}$ has to find z^* such that $\mathcal{D}(z^*)$ is statistically close to $\mathcal{T}$.

For the sake of contradiction, assume that **SampBQP** = **SampBPP**. Then, there exists a PPT algorithm $\mathcal{C}$ such that $\text{SD}(\mathcal{D}(z), \mathcal{C}(z))$ is small for all z . Hence, given samples $x_1, \dots, x_t$ from $\mathcal{T}$, it is sufficient to find z^* such that $\mathcal{C}(z^*)$ is statistically close to $\mathcal{T}$, which is agnostic *classical* distribution learning. From Theorem 1.11, it is possible with access to a Σ_3^{P} oracle.

⁶We do not know whether Σ_3^{P} is optimal. This could be improved.